CLASSIFICATION OF SOIL IMAGES USING CONVOLUTIONAL NEURAL NETWORK

MULTI-DISCIPLINARY PROJECT

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1. Abstract

Soil classification is an essential part of construction and civil engineering, as it directly affects decisions related to foundation design, site suitability, and structural safety. Understanding the type and properties of soil helps engineers determine how much load the ground can support and what kind of construction methods should be used. However, traditional methods of soil analysis are often slow, expensive, and require specialized knowledge and laboratory testing, making them less practical for quick decision-making on site.

This project presents a simplified and automated approach to soil classification using image processing and machine learning. The system uses a Convolutional Neural Network (CNN) based on the ResNet50 architecture to classify soil types from images. The model is trained on a labeled dataset of soil images and uses preprocessing techniques such as resizing, normalization, and data augmentation to improve its accuracy and performance.

To make the system practical and easy to use, it is deployed as a web application where users can upload soil images and instantly receive classification results. In addition to prediction, the application also includes features such as weather analysis and soil risk assessment, which help in evaluating site conditions more effectively.

Overall, the proposed system provides a faster, more accessible, and cost-effective alternative to traditional soil testing methods. It can assist engineers and construction professionals in making quicker and more informed decisions, ultimately contributing to safer and more efficient infrastructure development.

2. Introduction

Soil plays a critical role in construction, as every structure built on land depends on the strength and stability of the ground beneath it. Before starting any construction project, it is important to understand the type of soil present at the site, since different soils behave differently under load. For example, some soils can support heavy buildings, while others may lead to foundation failure, cracks, or long-term structural issues. Because of this, soil classification becomes one of the first and most important steps in civil engineering and site analysis.

Traditionally, soil classification is carried out through laboratory testing and field surveys conducted by experts. While these methods are accurate, they are also time-consuming, costly, and require specialized equipment and knowledge. In many practical situations, especially during early site inspection or small-scale projects, such detailed analysis may not always be feasible. This creates a need for a faster and more accessible method of identifying soil types.

With the advancement of technology, image processing and machine learning have opened new possibilities in automating such tasks. By analyzing visual features such as color, texture, and structure, it is now possible to classify soil types directly from images. Deep learning models, particularly Convolutional Neural Networks (CNNs), have shown strong performance in image-based classification problems.

In this project, a soil classification system is developed using a CNN model based on the ResNet50 architecture. The system is integrated into a web application that allows users to upload soil images and receive instant predictions. Additional features such as weather analysis and risk assessment further enhance its usefulness for real-world construction scenarios. This approach aims to provide a quicker, easier, and more practical solution for preliminary soil analysis in construction projects.

Figure 1 – Types of soil identified in the model

Objectives of the Study

The primary objective of this work is to design and develop an efficient and reliable deep learning-based system for automated soil classification using image data. The study aims to reduce dependency on manual soil analysis by leveraging advancements in computer vision and deep learning techniques.

The specific objectives of this work are as follows:

- To develop a robust image classification model capable of identifying different soil types based on visual features such as texture, color, and structure.
- To employ a transfer learning approach using the ResNet50 model in order to enhance feature extraction and improve classification accuracy.
- To classify soil images into five distinct categories: Alluvial, Black, Clay, Red, and yellow soil, ensuring accurate multi-class prediction.
- To preprocess and standardize input images through resizing, normalization, and transformation techniques to improve model performance and consistency.
- To evaluate the effectiveness of the proposed model using standard performance metrics such as accuracy, precision, recall, F1-score, confusion matrix, and ROC curve.
- To analyze classification results and identify potential sources of error, particularly in cases of visually similar soil types.
- To ensure that the developed system demonstrates good generalization capability and can be extended for real-world agricultural applications such as soil monitoring and crop planning.

3. Related Work

Soil classification has traditionally been carried out using physical and chemical analysis, which is time-consuming and requires expert knowledge. With advancements in image processing and machine learning, automated soil classification using images has gained significant attention.

Recent studies have shown that **Convolutional Neural Networks (CNNs)** are effective for soil classification. Singh et al. developed a CNN-based model using RGB images, achieving high accuracy, while Ahmad et al. proposed a deep CNN using UAV images, improving performance over traditional methods. To enhance accuracy, **transfer learning models** such as ResNet50 and InceptionV3 have been widely used. Pujari et al. demonstrated the effectiveness of ResNet50, and Hossain et al. showed that InceptionV3 captures complex soil features efficiently.

Lightweight models like **MobileNetV2** have also been explored. Maji and Chakraborty implemented MobileNetV2 for smartphone-based soil classification, achieving good accuracy with low computational cost. In addition, traditional methods such as **Support Vector Machines (SVM)** by Barman and Choudhury and other techniques like KNN and ANN have also been used for classification based on texture and color features.

Although these studies achieve good results, most focus only on classification. This project extends the approach by integrating a web-based system along with weather analysis and soil risk assessment for practical use in construction.

4. METHODOLOGY

4.1 Overview of Proposed System

The proposed methodology follows a structured pipeline for soil classification using deep learning techniques. The workflow includes dataset collection, preprocessing, feature extraction, model training, and performance evaluation. This systematic approach ensures accurate classification and efficient learning.

Like existing CNN-based frameworks, the model leverages transfer learning to enhance performance while reducing training complexity.

3.1 Dataset of Soil Images - The dataset consists of soil images categorized into five classes: **Alluvial, Black, Clay, Red, and yellow soil**. The images are collected under varying environmental conditions, including differences in lighting, texture, and background. The dataset has over ***ten thousand*** soil images which are

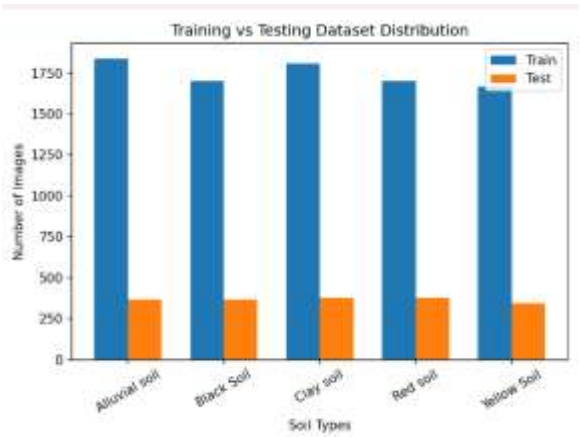

Figure 2 shows the class-wise distribution of soil images within the train and test dataset, comprising five distinct categories

uniformly spread across the soil types, this uniform distribution ensures class balance, which is a critical factor in training deep learning models for classification task The dataset is divided into:

- **Training set** for model learning
- **Testing set** for evaluation

This separation ensures unbiased performance assessment.

4.2 Image Processing and Preprocessing

4.2.1 Input Soil Images

The input data consists of raw soil images collected from different sources. These images are in RGB format and may vary in size, lighting conditions, and quality. Because of these variations, preprocessing is required before giving them to the model.

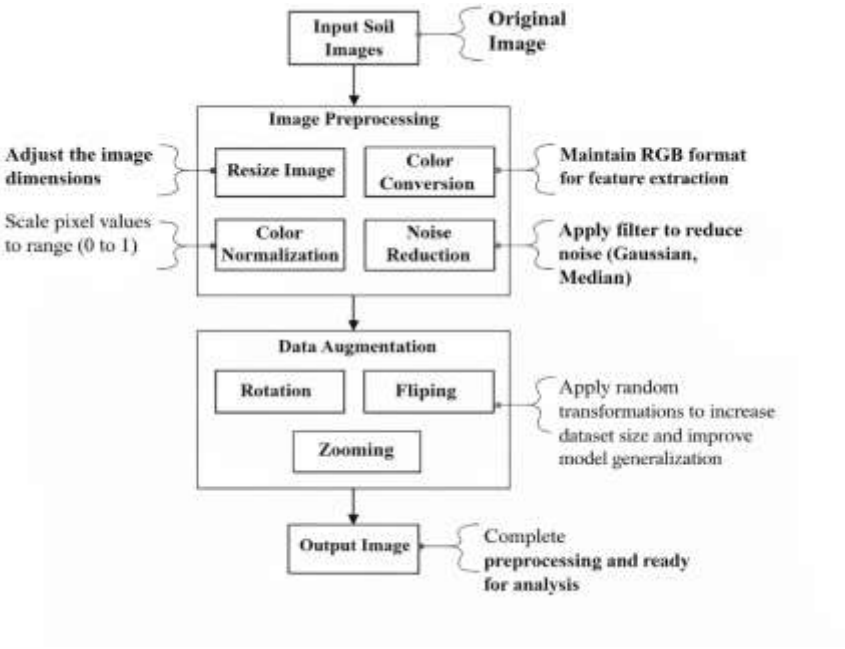

Figure 3. Flow Diagram of Image Preprocessing

4.2.2 Image Preprocessing

Image preprocessing is done to make the data consistent and improve the model performance.

a) Image Resizing:
All images are resized to **224 × 224 pixels** so that they match the input size required by the ResNet50 model. Neural networks generally need fixed input size to work properly.

b) Color Handling:
The images are kept in RGB format since color plays an important role in identifying different soil types.

c) Normalization:
Pixel values are scaled from **[0, 255] to [0, 1]**. This helps the model train faster and gives more stable results.

d) Noise Reduction:
Some basic noise reduction is applied to reduce unwanted variations like lighting changes and background noise, so the model can focus more on useful features.

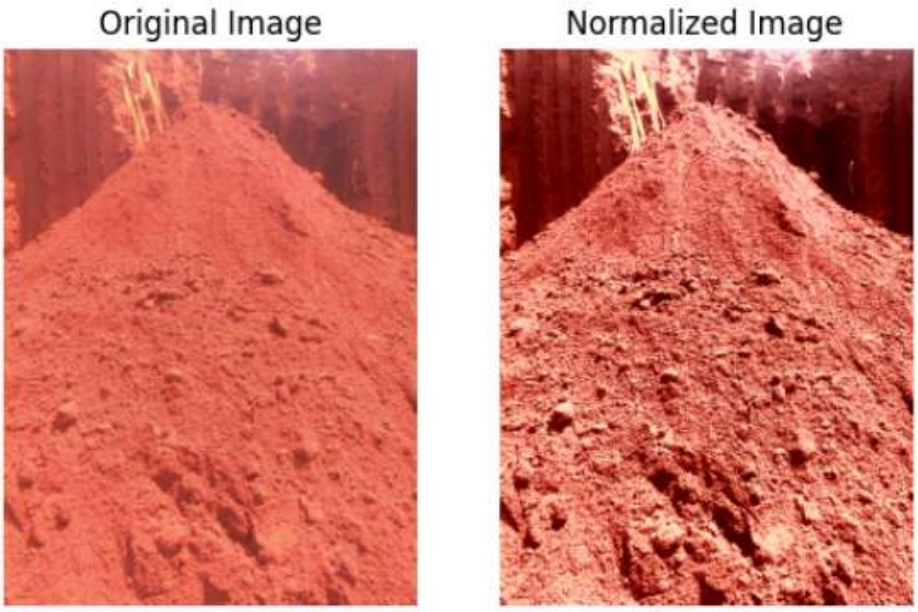

Figure 4 shows how input image is transformed and normalised

4.2.3 Data Augmentation

Data augmentation is used to increase the dataset size and make the model more robust.

a) Rotation:
Images are rotated at different angles so the model can learn from different orientations.

b) Flipping:
Horizontal and vertical flips are applied to avoid bias towards a particular direction.

c) Zooming:
Random zoom is used so the model can learn both full and close-up patterns of soil.

Purpose:
The main purpose of augmentation is to create more training data and help the model perform better in real-world conditions.

4.3 Deep Learning Model Architecture - ResNet50 Model

ResNet50 is a deep convolutional neural network architecture consisting of 50 layers, originally introduced in 2015. Its key feature is the use of residual learning through identity shortcut connections, which helps in overcoming the vanishing gradient problem and allows effective training of deeper networks. Due to its strong feature extraction capability, it has shown excellent performance in various image classification tasks and is widely used in transfer learning applications.

In this project, the pre-trained ResNet50 model is customized for soil classification. The original top classification layer is replaced with a new fully connected layer to match the required **five soil classes**: Alluvial, Black, Clay,

Red, and yellow soil. The modified model is then fine-tuned using the soil image dataset so that it can learn domain-specific features such as texture, color, and structure.

This approach helps in improving classification accuracy while reducing training time, as the model already contains pre-learned features from large-scale datasets.

Table 1. Architecture Overview of ResNet50

Layer/Block	Configuration/Action	Purpose
Base Model	ResNet50 (pre-trained on ImageNet)	Extracts generic features from soil images
include_top=False	Excludes original dense layers	Allows insertion of custom classification head
GlobalAveragePooling2D	Converts feature maps into 1D vector	Reduces dimensionality while retaining spatial information
Dense (1)	256 neurons, ReLU activation	Learns high-level features specific to soil images
Dropout (1)	Rate = 0.5	Regularization to prevent overfitting
Dense (Output)	Nine neurons, Softmax activation	Final layer for multi-class classification (nine soil classes)

4.4 Training Process

After the image processing, the training process is carried out using the training dataset. The preprocessed and augmented soil images are fed into the model in batches. During training, the model learns to extract important features such as texture, color, and patterns from different soil types.

The model is trained over multiple epochs, where in each epoch the entire dataset is passed through the network. A loss function is used to measure the difference between predicted and actual labels, and the optimizer updates the model weights to reduce this error.

The training performance is monitored using accuracy and loss values. As the training progresses, the accuracy increases while the loss decreases, indicating that the model is learning effectively.

Table 2. Training Parameter of ResNet50 Architecture

Parameter	Value	Description
Model	ResNet50	Pre-trained CNN used for feature extraction
Number of Classes	5	Alluvial, Black, Clay, Red, Yellow Soil
Input Size	224 × 224 × 3	Standard input size for ResNet50
Batch Size	32	Balance between memory usage and performance
Epochs	20	Number of training iterations

4.5 Evaluation Metrics

Evaluation was conducted using a multi-metric approach on the validation and test sets. This ensures that the models are not only accurate but also consistent and balanced across all classes.

Accuracy:
Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted samples to the total number of samples. It gives a general idea of how well the model performs but may not fully reflect class-wise performance.

$$Accuracy = \frac{TP+TN}{TP+FP+FN+TN}$$

Precision:
Precision indicates how many of the predicted positive samples are actually correct. It is useful in understanding the reliability of the model's predictions for each class.

$$Precision = \frac{TP}{TP+FP}$$

Recall:
Recall measures the ability of the model to correctly identify all actual positive samples. It shows how well the model captures all relevant instances of each soil type.

$$Recall = \frac{TP}{TP+FN}$$

F1-Score:
F1-score is the harmonic mean of precision and recall. It provides a balanced measure when there is an uneven class distribution or when both false positives and false negatives need to be considered.

$$F1\ Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where :

- TP: The number of correctly predicted positive classes.
- TN: The number of correctly predicted negative classes.
- FP: The number of incorrectly predicted positive classes.
- FN: The number of incorrectly predicted negative classes.

Confusion Matrix:
The confusion matrix provides a detailed class-wise analysis of model predictions. It shows the number of correct and incorrect predictions for each class, helping to identify misclassification patterns.

ROC Curve and AUC:
The Receiver Operating Characteristic (ROC) curve evaluates the model's ability to distinguish between classes. The Area Under the Curve (AUC) represents the overall performance, where values closer to 1 indicate better classification capability.

5. EVALUATION AND RESULTS

The evaluation of the soil classification models is an important step in assessing their effectiveness in accurately categorizing soil types based on input images. This step aims to provide a comprehensive analysis of the three employed models, leveraging key performance metrics such as accuracy and loss. The evaluation process involves analysing the learning behaviour and convergence of each model during training and validation. Through visual representations of accuracy and loss graphs, the study gain insights into the models' ability to generalize and make accurate predictions on both seen and unseen data.

5.1 Training Performance Analysis

Figure 4 shows two important training performance plots for a ResNet50 Model (CNN), one for accuracy and one for loss, across 20 epochs. Left plot is training and testing accuracy. In left plot Y-axis is accuracy and X-axis is training epoch. The training performance was evaluated using accuracy and loss curves over multiple epochs. The training accuracy increases from approximately **88% to 99%**, while validation accuracy improves from around **89% to 96–97%**, indicating effective learning

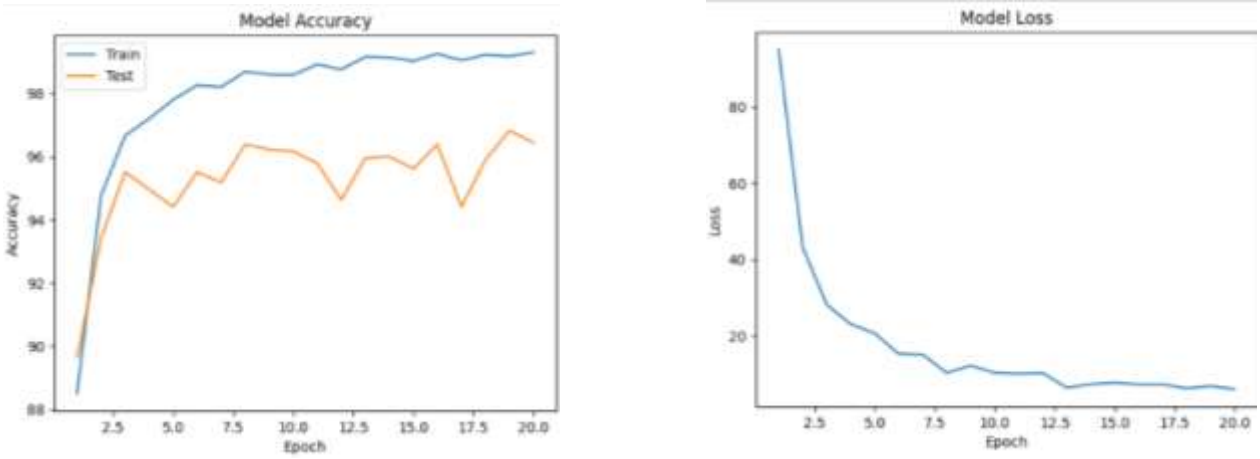

Fig. 5,6. ResNet50 Model Training Accuracy & Loss

Right plot of . ResNet50 Model shows training Loss and testing loss. In right plot Y-axis is training loss and X-axis is training epochThe loss decreases significantly from about **95 to below 20**, showing proper convergence. The small gap between training and validation accuracy suggests minimal overfitting and good generalization to unseen data.

5.2 Quantitative Performance Metrics

The performance of the model was further evaluated using precision, recall, and F1-score for each soil class. The overall accuracy achieved is **96%**, with strong and balanced metric values across all classes. Figure 5 shows the strengths and weaknesses of the model for each class

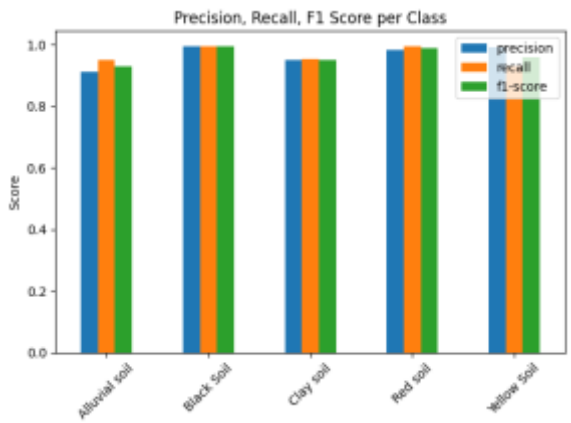

Fig. 7. Metrics evaluating the performance of a ResNet50 classification model

The model demonstrates strong classification performance, particularly for **Black and Red Soil (F1: 0.99)**, indicating highly accurate and reliable predictions. **Clay Soil (F1: 0.95)** maintains balanced performance, while **Alluvial Soil (F1: 0.93)** and **Yellow Soil (Recall: 0.93)** show minor limitations, suggesting occasional misclassification due to feature similarity. Overall, the average scores (~0.96–0.97) confirm that the model achieves consistent and well-generalized performance across all classes.

5.3 Confusion Matrix and Error Analysis

The confusion matrix provides a detailed representation of class-wise prediction outcomes by comparing actual and predicted labels. A strong concentration of values along the diagonal indicates that the majority of samples are correctly classified.

From the figure 6 , certain misclassification patterns are observed:

- **Clay soil is occasionally misclassified as Alluvial soil**
- **Yellow soil shows minor confusion with Alluvial soil**
- Other classes, particularly Black and Red soil, demonstrate near-perfect classification with negligible errors

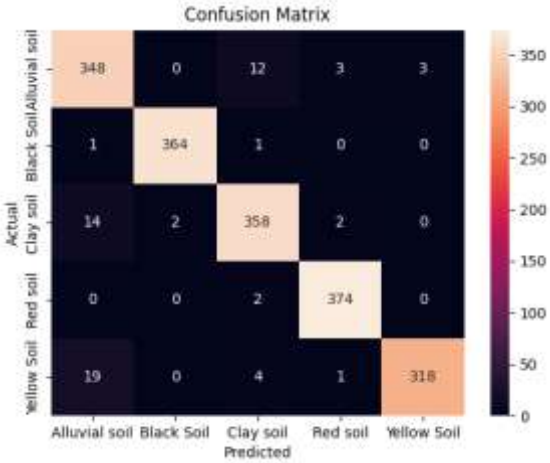

Figure 8 Confusion Matrix of proposed Soil Classification Model

These misclassifications can be attributed to the **visual similarity in soil texture, color distribution, and granularity**, which makes it challenging for the model to distinctly separate certain classes. Despite these minor overlaps, the overall error rate remains low, confirming the accuracy of the model.

5.4 ROC Curve and Discriminative Ability

The Receiver Operating Characteristic (ROC) curve was utilized to evaluate the model's ability to distinguish between different classes. A one-vs-rest approach was adopted for multi-class evaluation, and the Area Under the Curve (AUC) was computed for each class.

From the figure 5 ,we can see that most classes, including **Black and Red soil (AUC ≈ 1.00)**, achieve near-perfect separation, while **Clay soil (AUC ≈ 0.98)** shows slightly lower performance. The curves are concentrated near the top-left region, indicating high true positive rates and low false positive rates, thereby confirming strong classification reliability

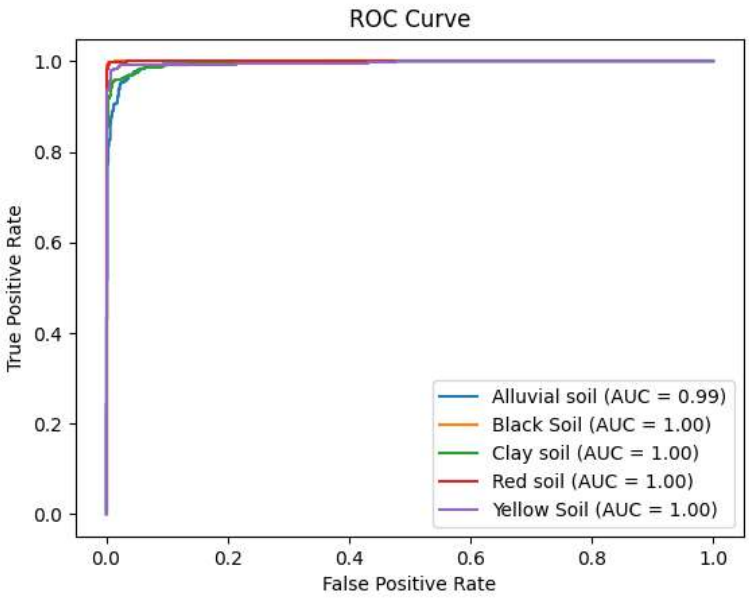

Fig. 9. ROC Curves for Multi-Class Soil Classification

6. CONCLUSION

To conclude, this project manages to create a reliable automatic soil classification system through image processing techniques and artificial intelligence. The customized soil image database, which comprises 10,000 images, representing various soil types, including Black, Alluvial, Red, yellow and Clay, was utilized in training and evaluating the efficiency of the classifier. Moreover, the model can deal with challenging circumstances, including different lighting, orientation, and background images, proving to be accurate, as it managed to classify soil types with ***96.50%*** success. The utilization of preprocessing steps, such as image resizing, normalization, augmentation, equalization, and filtering, has greatly influenced model performance.

In terms of implementation, the model is used in an interactive online website, allowing users to upload their soil images to the website and instantly receive information regarding the soil type. Moreover, such functions as weather analysis and soil risk assessments make the tool even more convenient and useful in construction activities. The proposed project offers faster access to soil data compared to traditional testing procedures, being cost-efficient and user-friendly. It is true that there exist several disadvantages related to image quality and size of datasets. Ongoing efforts to finetune hyperparameters and explore advanced neural network architectures aim to further improve the model's accuracy. Nonetheless, the developed system can serve as a basis for developing more advanced algorithms in the field.

7. FUTURE WORK

Improvements in the dataset can be done. Currently, the proposed system is based on a around 10 thousand number of soil images. In future studies, a larger and more diversified dataset may be created, which can include samples from various geographical locations under different lighting conditions.

Further improvement of the model should be related to advanced architectures of convolutional neural networks. Although ResNet50 architecture showed satisfactory results, other methods such as

EfficientNet or hybrid models can be used to further optimize the system performance and reduce the need for powerful computing resources.

The system can also be enhanced in terms of using camera phones to take soil pictures in real-time. As the user does not necessarily need to upload images but only identify the soil type, a mobile app can be designed that captures soil images automatically, thus providing increased usability.

The project may further evolve using geotechnical parameters in determining the type of soil. These parameters can be obtained via IoT devices and combined with the visual information gathered from soil images.

8. REFERENCES

[1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016.

[2] A. Krizhevsky, I. Sutskever, and G. Hinton, “ImageNet classification with deep convolutional neural networks,” in Proc. Adv. Neural Inf. Process. Syst. (NIPS), 2012.

[3] G. D. Chate and S. S. Bhamare, “Classification of soil images using convolutional neural network,” I.J. Image, Graphics and Signal Processing, vol. 17, no. 5, pp. 26–41, 2025.

[4] S. Dialani, “Soil classification through advanced image processing and deep learning models,” MSc Thesis, National College of Ireland, 2023.

[5] F. Chollet, “Xception: Deep learning with depthwise separable convolutions,” in Proc. CVPR, 2017.

[6] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. Chen, “MobileNetV2: Inverted residuals and linear bottlenecks,” in Proc. CVPR, 2018.

[7] C. Szegedy et al., “Rethinking the inception architecture for computer vision,” in Proc. CVPR, 2016.

[8] J. Deng et al., “ImageNet: A large-scale hierarchical image database,” in Proc. CVPR, 2009.

[9] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.

[10] S. Singh, A. Kumar, and R. K. Singh, “Soil classification using machine learning techniques,” Int. J. Eng. Res. Technol., vol. 9, no. 3, pp. 120–125, 2020.

[11] M. Ahmad et al., “Deep learning-based soil classification using UAV images,” Comput. Electron. Agric., vol. 158, pp. 123–135, 2019.

[12] P. Pujari, R. Yakkundimath, and A. Byadgi, “Image processing based detection of fungal diseases in plants,” Procedia Comput. Sci., vol. 46, pp. 1802–1808, 2015.

[13] U. Barman and R. D. Choudhury, “Soil texture classification using machine learning,” Int. J. Comput. Appl., vol. 178, no. 10, pp. 1–5, 2019.

[14] A. Krizhevsky, “Learning multiple layers of features from tiny images,” Univ. Toronto, 2009.

[15] T. K. Ho, “Random decision forests,” in Proc. Int. Conf. Document Analysis and Recognition, 1995.

[16] L. Breiman, “Random forests,” Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001.

[17] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in Proc. ICLR, 2015.

[18] R. Szeliski, Computer Vision: Algorithms and Applications. Springer, 2010.

[19] A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O’Reilly, 2019.

[20] S. Haykin, Neural Networks and Learning Machines. Pearson, 2009.